Eamon McGleenan

Data Scientist • Sports Performance Analytics, Machine Learning & Data Visualisation

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PROFILE

Data scientist and physicist finished PhD in Predictive Sports Analytics at Queen's University Belfast, in collaboration with STATSports. 6+ years Python experience, turned high-frequency data into validated metrics, forecasting models and dashboards. I built and run GPS data system for Gaelic football team, delivered demand-forecasting tool for a UK water utility as a freelancer, and developed new player-fatigue metrics that are now features in STATSports' elite-sport software. I use AI coding tools (Claude Code) every day and review their output as carefully as my own. Two first-author peer-reviewed publications, with a third under review.

KEY SKILLS

- **Programming & data:** 6+ years of Python (pandas, NumPy, scikit-learn, matplotlib, Plotly); SQL database design and querying, including window functions, CTEs and nested JSON data in Snowflake and BigQuery; advanced SQL analytics project in DuckDB; R; MATLAB; Excel; Git/GitHub; Linux command line.
- **Machine learning & statistics:** Random Forest, XGBoost, KNN, regression, classification and mixed-effects models; time-series forecasting with prediction intervals; anomaly detection; validation with cross-validation, walk-forward and out-of-time testing (ROC-AUC, MAE, RMSE, R^2, WAPE); reliability and minimum-data-requirement analysis.
- **Data engineering & delivery:** End-to-end pipelines from ingestion, cleaning, validation and gap-filling through modelling, reporting, export and scheduled refresh; packaging models as installable tools that clients run themselves.
- **Visualisation & reporting:** Tableau and Power BI dashboards; interactive web dashboards (Chart.js, Plotly) hosted on GitHub Pages; browser-based tools with client-side processing, pseudonymisation and encryption; information design for non-technical audiences.
- **Sports & sensor data:** STATSports Apex 10 Hz GPS/GNSS and Sonra; load, sprint, acceleration and fatigue metrics; Butterworth filtering, acceleration–speed profiling, power–force–velocity profiling, ballistic power, changepoint detection and wavelet analysis; FBref and Transfermarkt player data; wearable heart-rate and HRV data.
- **AI-assisted development:** Daily use of Claude Code for prototyping, coding, refactoring, visualisation and workflow automation, including agentic workflows, with careful review and testing of their output.
- **Communication & stakeholders:** Requirements gathering with clients; turning coach and practitioner questions into practical measures; live match-day feedback; national coaching-conference speaker; RTÉ News and The Irish Times features; university teaching.
- **Coaching, sport science & commercial:** GAA/LGFA Youth Athletic Development (Level 1) and Setanta College qualifications; plan and deliver pitch and gym sessions.

PROFESSIONAL EXPERIENCE

Freelance Machine Learning Engineer — Self-employed

2026–Present

- Gathered requirements with a UK water utility and built a 24-hour-ahead demand forecasting system for operational planning, handed over as a finished product that the client now runs independently.
- Compared Random Forest, XGBoost and KNN models on 15-minute data, validated on held-out, walk-forward data (WAPE, MAE, RMSE, R^2) and produced 80% prediction intervals so planners can see the uncertainty as well as the central estimate.
- Built the full pipeline: automated ingestion, cleaning, gap-filling, scheduled re-forecasting and export, with an anomaly-detection layer that flags data-quality and model issues before they reach users.
- Packaged the model as secure, platform-specific builds that install and run on the client's own machines.

GPS Data Analyst — Senior Footballers, O'Connell's Tullysaran GAC

2025–Present

Supervisor: Pauric McGlone

- Built and maintain an end-to-end Python and Tableau system that downloads, cleans, validates, analyses and visualises STATSports Apex data for every first- and reserve-team session and match.
- Designed the dashboards and weekly player reports that management, coaching and S&C staff use for load, availability and selection decisions; give live feedback during training and matches in plain language.
- Start from the decision a coach needs to make, agree the few measures that answer it, then refine each report on staff feedback; document how every metric is defined.

- Analysed three seasons (2024–2026) of match data across rule changes and promotion, resolving data-quality issues such as implausible durations and effort-detection artefacts (first-author paper under review).
- Part of the data-led programme that took the club to senior status for the first time in its 135-year history, featured by RTÉ News and The Irish Times.

U18s Performance Coach — O'Connell's Tullysaran GAC *2025–Present*

Supervisor: Fiona Donnelly

- Plan and deliver pitch-based conditioning, skills and gym-based strength sessions, tailoring load to players at different stages of growth and maturation.
- Monitor load and safeguard players across underage squads, working closely with the wider coaching staff.

PhD Researcher, Predictive Sports Analytics — Queen's University Belfast *2023–Present*

In collaboration with STATSports · Supervisor: Prof. David Jess

- Three years analysing large, high-frequency (10 Hz) tracking datasets from training and matches: cleaning, quality checks, modelling, analysis and visual reporting.
- Developed and validated new physical-performance and fatigue metrics (in-situ power–force–velocity profiling, ballistic power) from normal training and match data rather than separate testing, combining physics-based models with mixed-effects and time-series methods.
- Validated the metrics against criterion measures and established how much data each needs to be reliable, so practitioners can use them in real time.
- Worked with STATSports staff to turn the research into features in their elite-sport software; now developing curvilinear running metrics from raw tracking exports.
- Two first-author peer-reviewed publications and a third under review; presented to practitioners at the GAA & LGFA Coaching Conference.

Mathematics Demonstrator — Queen's University Belfast *2024–Present*

Supervisor: Prof. Ernst De Mooij

- Deliver tutorials to Physics BA and MSc students, working through assignment questions and explaining complex quantitative material clearly, with weekly graded feedback to deadline.

Placement Software Engineer — Global Automation *2021–2022*

Supervisor: Brian Mullin

- Designed and queried SQL databases recording wastewater-treatment site data, and built Power BI dashboards for governmental operational reporting.
- Supported software development, commissioning, testing and on-site deployment as part of a delivery team; developed and tested PLC programs and HMI interfaces.

SELECTED PROJECTS

Football Player Valuation & Trajectory Model *2026*

- Joined FBref and Transfermarkt data into ~70,000 player-seasons (2012–2025) and modelled which players' careers are on the rise (gradient boosting; out-of-time ROC-AUC 0.81, 3.1× top-decile lift).
- Published a transparent, plain-language scouting screen of ~6,700 young players across Europe's ten strongest leagues, scored on performance, intensity, misprofiling and underuse within positional peer groups (back-test hit rate 55% against a 33% baseline).

GPS Performance Dashboard — Session Analysis & Load Monitoring *Open source*

- Browser-based system that turns raw 10 Hz STATSports exports into session dashboards at 10-second resolution, flags the most demanding passages of play and tracks seasonal load with changepoint detection and wavelet analysis; pseudonymisation and in-browser encryption built in.

In-Situ PFv Profiling — Interactive Practitioner's Guide *2026*

- Interactive companion to my PFv research that guides practitioners through filtering, acceleration–speed profiling and in-match fatigue monitoring, processing STATSports files locally with nothing uploaded.

Athlete Monitoring Dashboard *2026*

- Python data pipeline feeding an interactive Chart.js dashboard on GitHub Pages that tracks acute:chronic workload ratio, resting heart rate and HRV trends over time.

Advanced SQL Baseball Analysis *In progress*

- Self-directed SQL project on MLB data in DuckDB, practising joins, window functions, CTEs and aggregations on a real multi-table dataset.

EDUCATION

PhD, Predictive Sports Analytics — Queen's University Belfast *2023–2026*

Supervisor: Prof. David Jess

- CAST studentship in collaboration with STATSports (market leaders in GPS tracking for elite sport); completing October 2026. Research: in-situ fatigue analysis, power–force–velocity profiling, data analysis and reporting.

MPhy Medical Physics (2:1, Integrated Master's) — Queen's University Belfast *2018–2023*

- Strong grounding in mathematics, physics, statistics and programming. Final-year project: statistical insights into player and team metrics from STATSports sensors, applying astrophysics tools to sports-technology data.

A-Levels: AAB (Mathematics, Physics, Biology) — St Patrick's Grammar School, Armagh *2011–2018*

CERTIFICATIONS

- Hands-On Essentials: Data Warehousing Workshop badge – Snowflake (2026).
- Advanced SQL – Kaggle (2026).
- Machine Learning Crash Course: Classification – Google Developer Experts (2026).
- Machine Learning Crash Course: Datasets, generalization, and overfitting – Google Developer Experts (2026).
- F3 Youth Athletic Development Level 1 — GAA / LGFA / Camogie (2026).
- Introduction to Machine Learning Using MATLAB — Queen's University Belfast (2024).
- Setanta College — Explosive Rotational Power; Personal Guide to Health & Fitness (2024).

PUBLICATIONS, TALKS & MEDIA

- McGleenan et al. (2026). Ballistic power: GPS-derived measurements suitable for the real-time monitoring of neuromuscular fatigue during repeated sprints. International Journal of Sports Science & Coaching. doi:10.1177/17479541251413077 (Full Text).
- McGleenan et al. (2026). In-situ PFv profiling in sub-elite Gaelic football: validation, comparison with isolated sprints, minimum data load requirements, and exploring "real-time" monitoring potential. International Journal of Sports Science & Coaching. doi:10.1177/17479541261468202 (Full Text).
- McGleenan et al. Match running demands in club Gaelic football across three seasons spanning rule amendments and promotion: a single-club GNSS case study. (Under review).
- Speaker (Nov 2025): Applying GPS Technology to Gaelic Games — Evidence-Informed Recommendations for the Weekly Microcycle. GAA & LGFA Coaching Conference, Newforge, Belfast.
- Media (Sept 2025): RTÉ News — "Big data helps GAA club to first senior status"; The Irish Times — "GAA moneyball"; The Sideline Eye.

BEYOND WORK

From a rural County Armagh farming family, with a strong work ethic. Lifelong Gaelic footballer and underage coach with a love of field sports in all codes; marathon and 15km road-race runner with a dedicated strength-training routine.